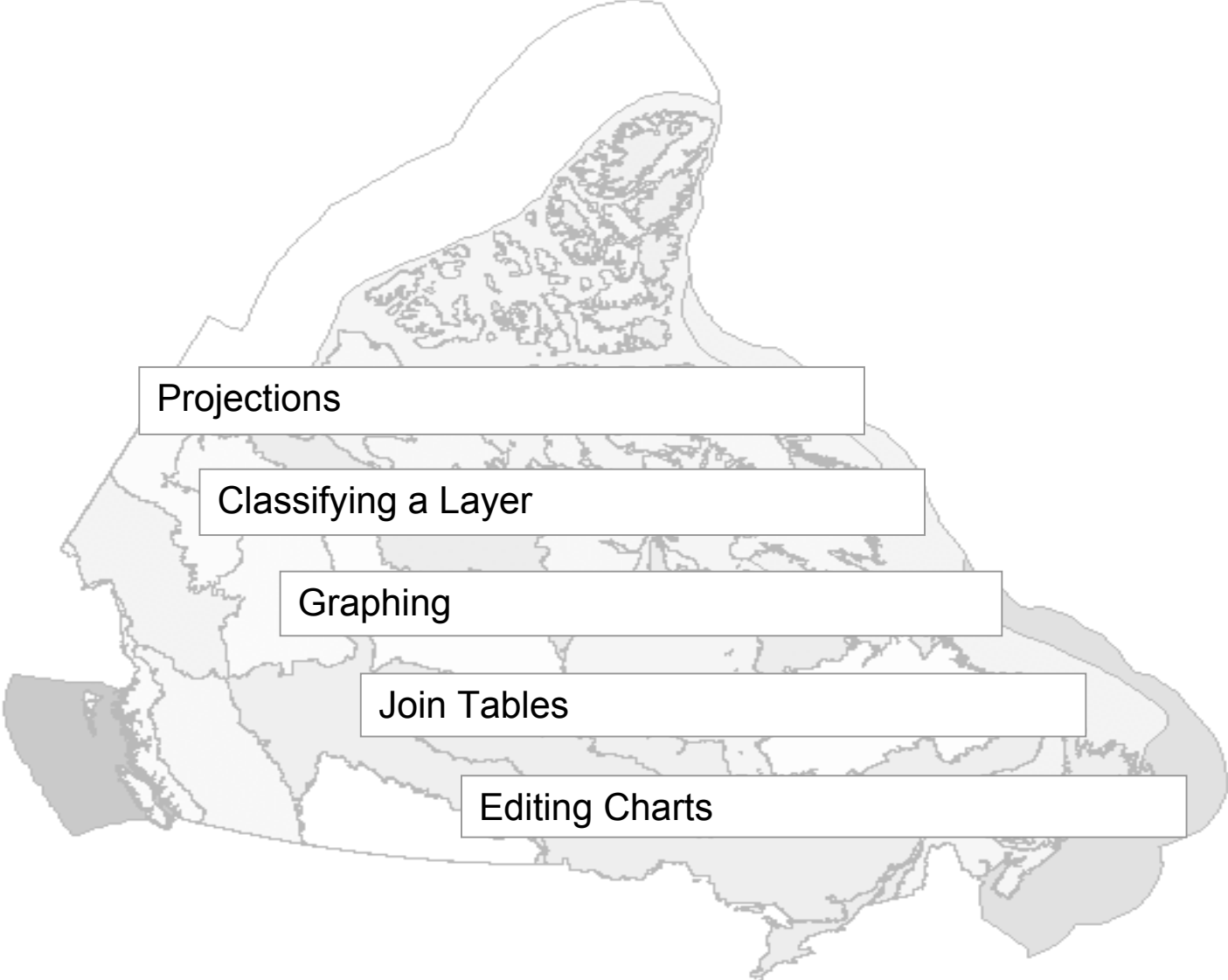


# **Ecozones of Canada**



Projections

Classifying a Layer

Graphing

Join Tables

Editing Charts

## GIS Information

### **Here's a list and explanation of important features in ArcView.**

**Project :** In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

**View**  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

**Table of Contents:** All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

**Theme:** A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

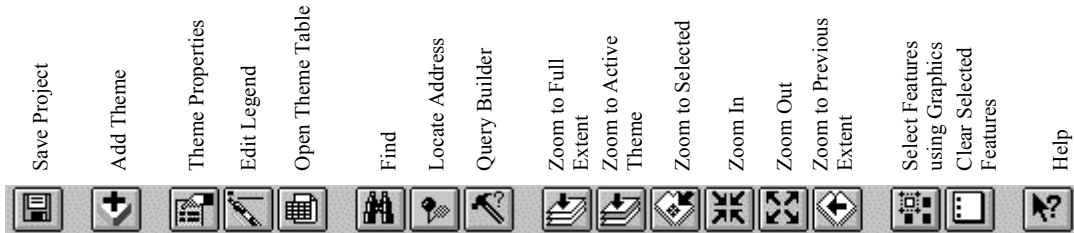
**Tables**  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

**Layout**  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

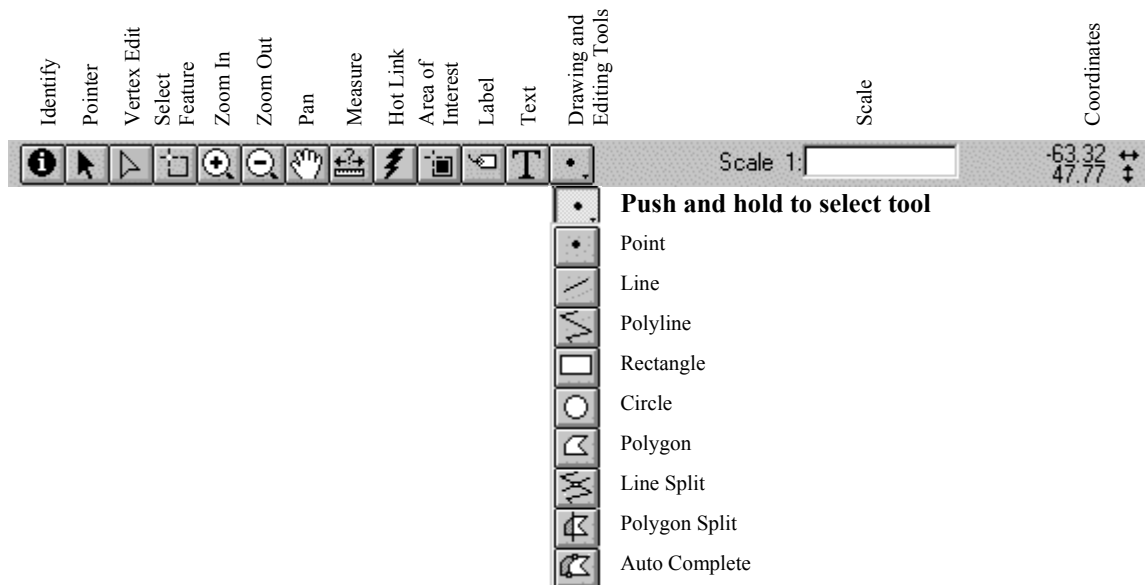
**Hotlink:** In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

## ArcView Buttons Reference Guide

### View - Buttons



### View – Tools



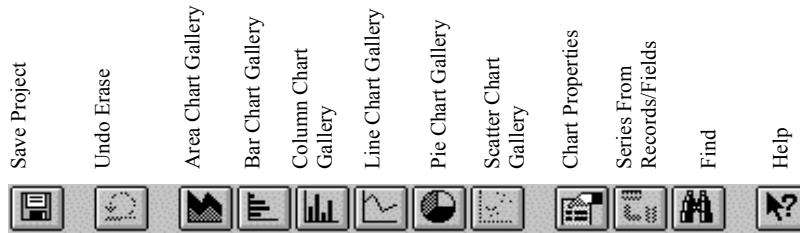
### Table - Buttons



### Table - Tools



## Chart - Buttons



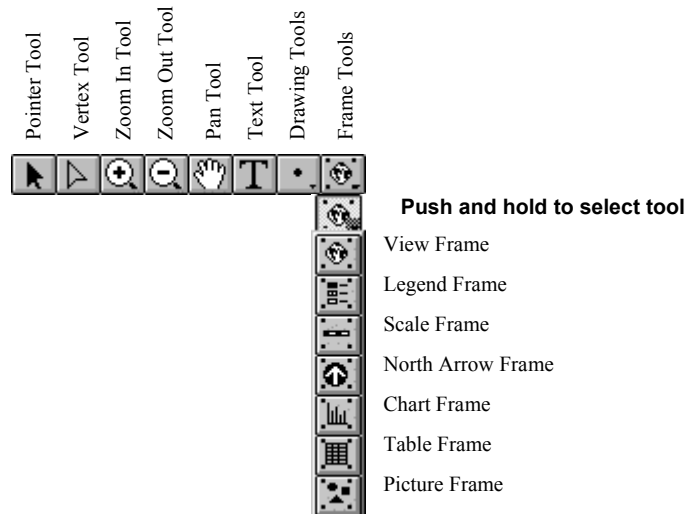
## Chart - Tools



## Layout - Buttons



## Layout - Tools



For more Information: <http://www.esricanada.com>

## Getting Started

**Before you begin**, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

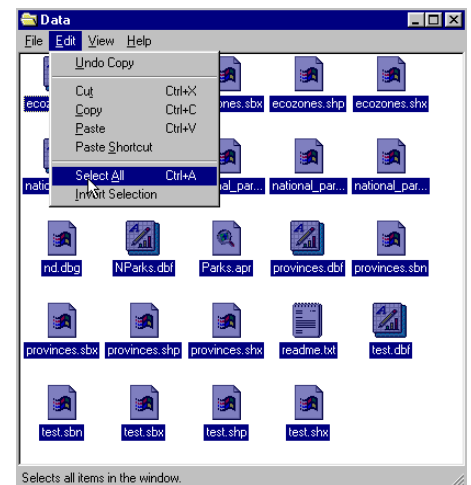
**!** (The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).

To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /**.

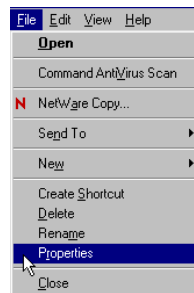
Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For windows based systems follow these instructions:

**!** (The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)

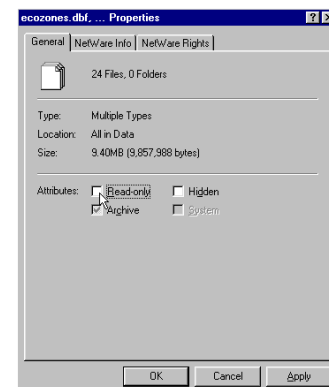
1. Open your new data folder
2. Choose **Edit > Select all** (they will all appear Blue)



3. From the menu choose:  
**File > Properties**



4. Remove the check mark in the box beside the label **read only**.



5. Click **apply**, then **OK**

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

**!** Denotes an observation.

**?** Denotes a question to be answered on the Student Activity Sheet.

## STARTING THE PROJECT

The purpose of this lesson is to learn about map projections, map classification, data graphing, table functions and map presentation while studying Canada's Ecozones.

### *STARTING ArcView*

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **O.K.**



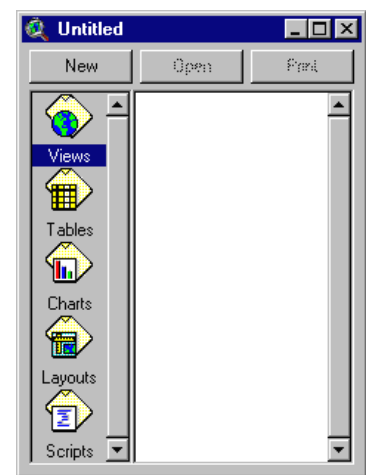
When the program starts, it automatically opens a project window titled "**Untitled.**"

If you like, you can maximize the **ArcView** window to fill the screen.

*In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.*

3. Now you are ready to add documents to your new project.

4. With the "**Views**" icon selected, click on **New** and maximize the **View1** window.

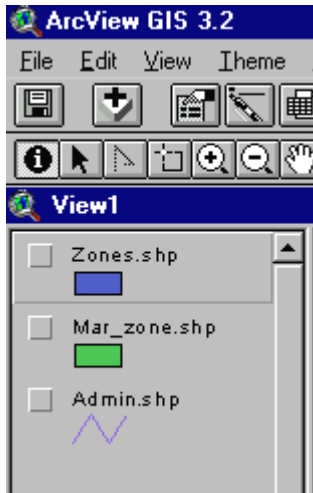


5. Click the **Add theme button**  and Navigate to the data folder where your data is stored. Hold down your SHIFT key and select the following files:

*Zones.shp*  
*Mar\_zone.shp*  
*Admin.shp*

6. Click **O.K.**

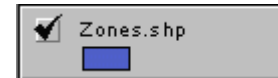
This will add these three themes to the table of contents of the new view. They are now available for display and editing.



7. Click and drag the Admin.shp theme up to the top of the table of content if necessary.

! Note the themes are displayed from the bottom to the top so line themes should be on top of polygon files to be visible.

8. Place a check mark next to the Zones theme by clicking on the box next to it. This will make it visible



9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **apr.** means that you are saving your work as a project that can be opened up in the future)

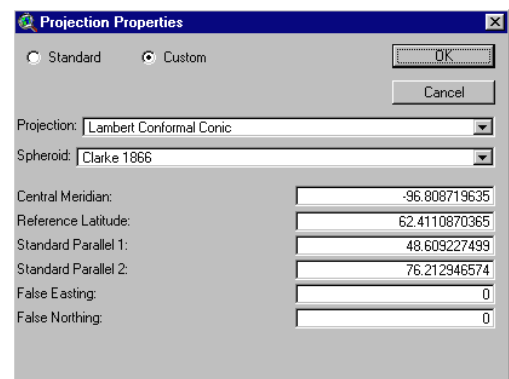
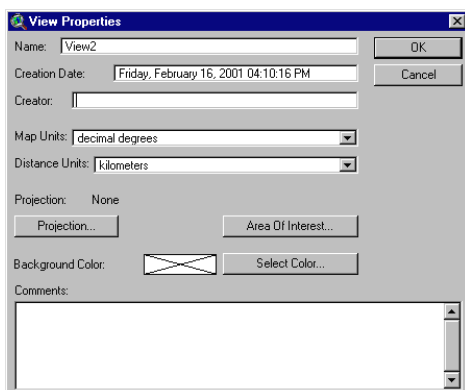
## ArcView - Changing the Map Projection

! A **Map Projection** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

### CHANGING PROJECTION

Now we will give our map “projection coordinates” to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties**.
3. Select **Map Units** list. Change the **map units to decimal degrees**.
4. Select **Distance Units** list. Change the **distance units to kilometers**
5. Click the **Projection** button.
6. Select **Custom**.
7. Select **Lambert Conformal Conic** from the projection list.



**!** Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made.

Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles).

**Conformal** implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.

**Conical** means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The **central meridian**, **false easting** and **false northing** are used to define the X (easting) and Y (northing) coordinates of the map. The **reference latitude** and **standard parallel 1 and 2** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

8. Click **O.K.**

The bottom of the **View Properties** window should look like this:



9. Click **O.K.**

**!** Note the difference in the map presentation.

**?** How is it different from the non-referenced map?

## ArcView - Working with layers

### ADDING POLITICAL BOUNDARIES

1. Check mark the **Admin Theme**  to make it visible.

### VIEWING A LAYER

**!** We want to emphasize the individual ecozones in this new projection, Lambert Conformal Conic projection.

1. Double click **Zones.shp** layer in the list of themes, which will take you to **Legend Editor**.

2. Open **Legend Type** list.

3. Select **Unique Value**.

4. Open **Value Fields** list.

5. Select **Name**.

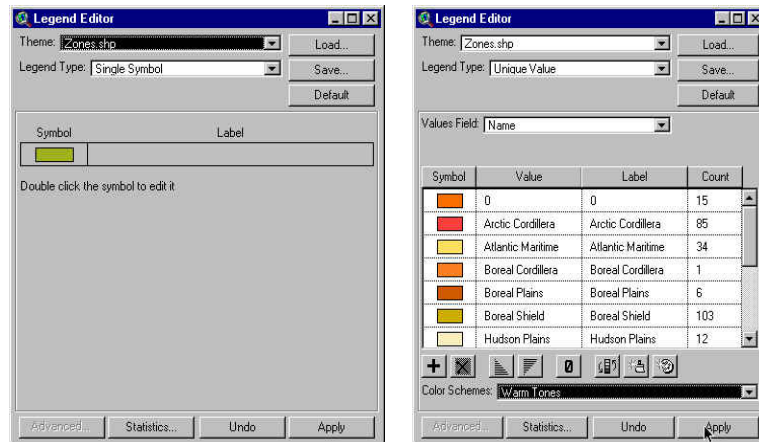
! This will display a list of the 15 terrestrial Ecozones with legend colours.

6. Now select **Colour Schemes** list.

7. Select a different **Color scheme**. e.g. **Warm Tones**

8. Click **Apply**.

! This will now display the Ecozones with a different set of colours.



## CREATING A THEMATIC LAYER

You will now create a thematic map. This will require classifying the Ecozones based on their permafrost content. This Classification will then be used to make a map of permafrost concentration.

1. **Uncheck** the **zones.shp** layer to turn off the visibility.

2. Click **Add theme**  button.

3. Select **zones.shp** to add a second zones theme to the current view.

4. Check new **zones** layer box. (Choose the new one with the single color legend)

5. Double click the new zones layer.

6. Select **Unique Value** legend type.

7. Select **pf-code** values field which is a code for the percent permafrost found in each ecozone (see the table below).

Edit the legend to reflect the percentage of permafrost in the class. (Change labels as shown)

| VALUE | LABEL                                        |
|-------|----------------------------------------------|
| CI    | >90% - Continuous permafrost:                |
| Elm   | >50-90% - Extensive discontinuous permafrost |
| SI    | >0-50% - Sporadic discontinuous permafrost   |
| N     | 0% - No permafrost                           |

8. Click **Apply** and close **Legend Editor**.

9. Click and drag **admin** layer to the top of the theme list if necessary.

## COMPARING LAYERS

Now that we have a permafrost map lets make a forest map and compare them.

1. Click **Add theme**  button.
2. Select **zones.shp**, to add a third zones theme to the current view.
3. Click **O.K.**
4. Double click the new **zones** layer.
5. Choose **Legend Type** *unique value*.
6. Select **Value Field** *forest*.
7. Click **Apply** and close **Legend Editor**.
8. Compare this new forest map with the permafrost map.

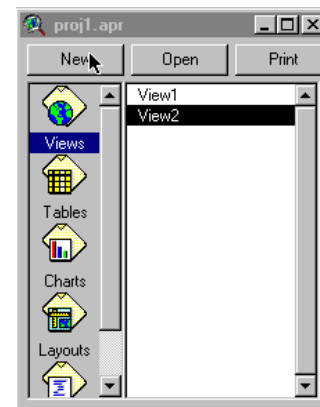
? Compare the extent of the permafrost and the forested area by checking the box to open and close layers. Is there any similarities or differences between the forested areas and the permafrost areas?

? Are the Prairie and Mixedwood Plains ecozones to the south of the permafrost areas?

? Why are the Prairie and Mixedwood Plains non-forested?

## ArcView - Data Graphing

We have produced some maps in this lesson. However, there is more we can do with the information we have about these maps in table format. The coastline of Canada can be relatively simple as it is along the St. Lawrence Gulf coast or extremely complex as it is along the Pacific or Labrador coastlines. One table gives us the length of each of the **marine** ecozones. Let's compare these values in a bar graph.



### DISPLAY CUSTOMIZED MAP

1. First create a New View from your project window "view 2".
2. Click **Add Theme** .
3. Add **Mar\_zone.shp**.
4. Double click **Mar\_zone.shp** to open **Legend Editor**.

5. Change **Legend Type** to **Unique Value**.
6. Change Value Field to **marine**.

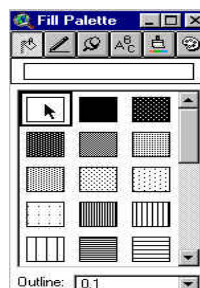
7. Select the color scheme **The High Seas**.
8. Double click the colour symbol **Land**.

Values Field: Marine

| Symbol                                                                              | Value              | Label              | Count |
|-------------------------------------------------------------------------------------|--------------------|--------------------|-------|
|  | Arctic Archipelago | Arctic Archipelago | 3     |
|  | Arctic Basin       | Arctic Basin       | 1     |
|  | Atlantic Marine    | Atlantic Marine    | 10    |
|  | Land               | Land               | 1366  |
|  | Northwest Atlantic | Northwest Atlantic | 13    |
|  | Pacific Marine     | Pacific Marine     | 3     |

Color Schemes: The High Seas

9. Select **clear fill pattern**.
10. Click **Apply**.
11. Close **Legend Editor**.



## JOIN TABLES

1. Return to your Project Window by selecting **window** in the main menu and selecting the **Untitled** window from the list of windows.
2. Select **Tables icon**.
3. Click **Add**.
4. Select **mar\_zone.dbf** from the data folder.
5. Click **O.K.**

Also open the Table named **sum1.dbf**.

Position the tables so you can see them both on the screen.

6. Select the column **Marine** in the **mar\_zones.dbf** table.
7. Now select the **Ecozone** column from the **sum1.dbf** table.



Make sure **sum1.dbf** is the window selected before you click join table.

8. Now click the **Join Tables** button. 

You now have one table with the information from both tables.

## CREATING A GRAPH

1. Click on and Select **Land** under the Ecozone column of the new combined table. This record (**land**) will be highlighted in yellow.
2. Click **Switch Selection** button  to reverse the selected records. Now all records except land are highlighted in yellow.
3. Click **Create Chart** button.  to create a chart using the highlighted records.
4. Under **Fields**, select **Coast**.
5. Click **Add**.
6. Open **Label Series Using** list.
7. Select **ecozones**.
8. Click **O.K.**

## EDIT THE CHART

1. Click the **Chart Element Properties Tool** button. 
2. Click on **Title** in the chart.
3. Change the name to **Coastal Length**.
4. Click **O.K.**
5. Click on the **y-axis**.
6. Check the **Axis Label** box
7. Change the label to **Kms**.
8. Click **O.K.**
9. Click on the **x-axis**.
10. Change the name to **Ecozones** in the **group label**.

11. Click **O.K.**

? Viewing the chart that is created. Which marine ecozone has the longest coastline?

? Which marine ecozone does not touch the coastline of Canada?

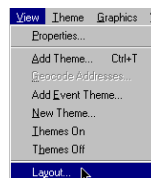
! Try changing other features on your graph through the **Table Properties** button.

## ArcView - Map Layout for printing

Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

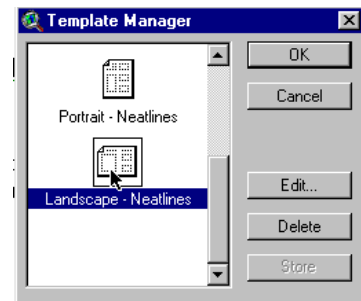
### SET UP MAP

1. Click Menu **view**.
2. Click **Layout**.



3. Select the **layout template** you want to use  
(One of the landscape templates is suggested) .

4. Click **O.K.**
5. Maximize the Layout screen.



### EDIT TITLE

1. Double click on the title **View1**.
2. Edit the title to reflect the purpose of the map, e.g., forestry, population.
3. Click **O.K.**
4. Resize the title using the corner handles and move it into a position that is visually balanced.
5. Reposition the legend if necessary.

### EDIT NORTH ARROW

1. Double click on the *north arrow* and select a different presentation. Ensure that that the arrow is pointing in the representative north direction of the map.
2. Reposition or rotate (using rotation function) as needed.

### EDIT SCALE BAR

1. Double click the *scale bar*.
2. Choose a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. Change the units to **kms**.
4. Find a combination of style, scale and divisions that is visually pleasing
5. Position to an appropriate location.

### ADD LABELS, TEXT, OTHER INFORMATION

1. Click on the **Text** tool.
2. Click at the position on your layout where you want the text to appear.



3. In the **Text Properties** box, enter your text, e.g., “produced by (your name, date)”.
4. Select desired formatting information such as **alignment, spacing and rotation**.
5. Click O.K. when it appears, as you want it.
6. Click on **Pointer** tool.
7. Select the new text
8. Click once to change location or size, Click twice to change content.

**!** Experiment with all of the components of the layout until you have the look you want.

### **CHANGE THE NAME OF THE LAYOUT**

You only have to do this if you want to save your project on the computer or on a disk.

1. Click menu **Layout**.
2. Select **Properties**.
3. Change the “**name**”. Click **O.K.**

### **ARRANGING YOUR LAYOUT**

- !** Experiment with the layout buttons (use the tools page provided at the beginning of this lesson)
- to **group** (tie pieces together so they move as one)
  - to **ungroup** (break apart the ties so that individual pieces move)
  - **bring to front** ( change order that pieces were laid down, e.g, lines on top of polygons)
  - **send to back** ( change order that pieces were laid down, e.g, polygons behind lines)
  - **zoom** ( make image bigger or smaller)
  - **neat line** (border around layout)

### **PRINT MAP LAYOUT**

1. Click **Print** button.
2. Click **Setup**.
3. Select your printer properties as outlined by your teacher.
4. Click **O.K.** to print.

**! Note** that the map can also be saved to a printer file.

**?** Include a printed version of your assignment. The layout should include a view of Canada and a chart. Remember to have a title with a purpose.

**?** List your title or purpose of your map here.

**?** Who would use, and how would these people use your map?